

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2023/2024

MAY/JUNE EXAMINATION

BACS2163 SOFTWARE ENGINEERING

TUESDAY, 4 JUNE 2024

TIME: 6.30 PM – 8.30 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS) IN DATA SCIENCE
BACHELOR OF COMPUTER SCIENCE (HONOURS) IN INTERACTIVE SOFTWARE
TECHNOLOGY

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

BACHELOR OF SOFTWARE ENGINEERING (HONOURS)

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

BACS2163 SOFTWARE ENGINEERING**Question 1**

- a) M2K is interested in creating an educational system based on the four key layers of layered software engineering technology. Describe each layer based on the system. (12 marks)
- b) Ram is a fresh graduate from the School of Computer Science. Recently he is hired by a software house to develop bespoke software for clients. Describe **THREE (3)** important code of ethics that Ram needs to follow as a software engineer in the software house. (9 marks)
- c) Discuss the **TWO (2)** software development issues that today's software engineers are likely to encounter. (4 marks)

[Total: 25 marks]

Question 2

- a) TLE Solutions is a company that develops mobile applications, with a focus on banking applications. Their job was to develop an application for ABC Bank. Thus, TLE used viewpoint-oriented analysis to make sure that the application requirements were understood comprehensively.
- (i) Identify any **TWO (2)** advantages and **TWO (2)** disadvantages of using viewpoint-oriented analysis method in requirements gathering and analysis. (4 marks)
- (ii) Identify any **SIX (6)** viewpoints of ABC Bank application and prepare a viewpoint hierarchy. (12 marks)
- b) In a project management, a project manager would need to consider factors that could motivate employees throughout the project. Describe any **THREE (3)** levels of needs in Maslow's motivation model. (9 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 3**

- a) Answer the following questions based on given conversation shown in **Figure 2**.



Source: <https://www.google.com/search+process+model>

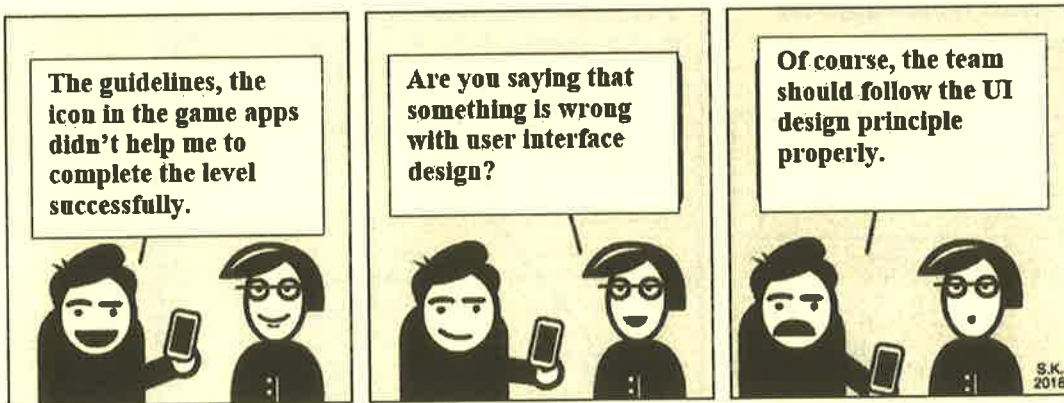
Figure 2: Conversation

- (i) Analyse the scenario above and suggest any **ONE (1)** suitable software process model to complete the above project. You may state any relevant assumptions to support your answer. (7 marks)
- (ii) Illustrate and explain the phases in above suggested software process model. (12 marks)
- (iii) Outline any **SIX (6)** of the Agile Software Development principles. (6 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 4**

- a) Based on given conversation shown in **Figure 3**, suggest and explain **SIX (6)** user interface designs principles that must be included in the mobile game application. (12 marks)



Source: <https://www.google.com/search?q=mobile+game+kids>

Figure 3: Conversation

- b) TPS Company is currently researching into Bank Security Management System (BSMS) application that offers an efficient and effective bank security management system. TPS allows customer to check their individual bank locker by providing CCTV access. Customer can also view the history access log time. In the event of unauthorised access, a notification message will be sent to police station automatically and to the customer at the same time.

- (i) Identify any **THREE (3)** non-functional requirements for the Bank Security Management System (BSMS) system. (9 marks)

- (ii) Explain **TWO (2)** challenges of user requirements gathering using natural language. (4 marks)

[Total: 25 marks]